

5 The mitotic cell cycle

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Interphase includes DNA replication before mitosis. Mitosis separates identical sister chromatids so daughter nuclei retain chromosome number and genetic identity, subject to mutation. Cell cycle checkpoints restrict division after DNA damage; uncontrolled proliferation can lead to tumours. A mitotic index estimates the fraction of observed cells undergoing mitosis.

Worked example: follow the reasoning

Of 200 root-tip cells, 24 are visibly in mitosis, giving mitotic index $24/200 = 0.12$, or 12%.

Quick check · multiple choice

A What separates to opposite poles during mitosis?

[1]

- A Sister chromatids
- B Homologous chromosomes only
- C Cell walls
- D Two intact nuclei

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A field has 18 dividing cells among 150 cells. Calculate mitotic index and explain why daughter cells have equal chromosome number.

[3]

Final answer: _____

Answers and explanation

Quick check: A. Sister chromatids

Question 1 [3 marks]

Index = $18/150 = 0.12$ [1]. DNA replicated in interphase [1], then sister chromatids separate to opposite poles [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.